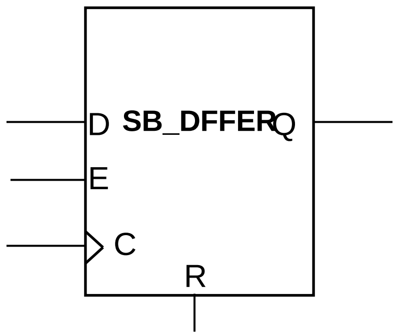
SB_DFFER

D Flip-Flop with Clock Enable and Asynchronous Reset

Data: D is loaded into the flip-flop when Reset R is low and Clock Enable E is high during a rising clock edge transition.

Reset: R input is active high, overrides all other inputs and asynchronously resets the Q output.



Inputs				Output
R	**E**	**D**	**C**	**Q**
1	X	X	X	0
0	0	X	X	Previous **Q**
0	1	0	↑	0
0	1	1	↑	1
Power on State	X	X	X	0

Key
↑ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input R: Logic '0'
Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF primitive without a Clock Enable port be used.

Verilog Instantiation

// SB_DFFER - D Flip-Flop, Reset is asynchronously on rising clock edge with Clock Enable.

```
SB_DFFER SB_DFFER_inst (
      .Q(Q),            // Registered Output
      .C(C),            // Clock
      .E(E),            // Clock Enable
      .D(D),            // Data
      .R(R)             // Asynchronously Reset
      );
```

// End of SB_DFFER instantiation

VHDL Instantiation

-- SB_DFFER - D Flip-Flop, Reset is asynchronously
-- on rising clock edge with Clock Enable.

```
SB_DFFER_inst : SB_DFFER
      port map (
      Q => Q,     -- Registered Output
      C => C,     -- Clock
      E => E,           -- Clock Enable
      D => D,     -- Data
      R => R            -- Asynchronously Reset
      );
```

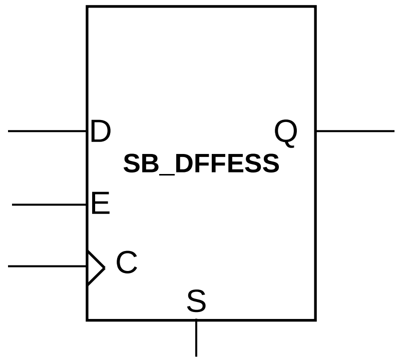
-- End of SB_DFFER instantiation

SB_DFFESS

D Flip-Flop with Clock Enable and Synchronous Set

Data: D is loaded into the flip-flop when S is low and E is high during a rising clock edge transition.

Set: Asserting S when Clock Enable E is high, synchronously sets the Q output.



Inputs				Output
S	**E**	**D**	**C**	**Q**
1	1	X	↗	1
0	0	X	X	Previous **Q**
0	1	0	↗	0
0	1	1	↗	1
Power on State	X	X	X	0

Key
↗ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input R: Logic '0'
Input S: Logic '0'

Verilog Instantiation

// SB_DFFESS - D Flip-Flop, Set is synchronous with rising clock edge and Clock Enable.

```
SB_DFFESS  SB_DFFESS_inst (
      .Q(Q),            // Registered Output
      .C(C),            // Clock
      .E(E),            // Clock Enable
      .D(D),            // Data
      .S(S)             // Synchronously Set
      );
```

// End of SB_DFFESS instantiation

VHDL Instantiation

-- SB_DFFESS - D Flip-Flop, Set is synchronous with rising clock edge and Clock Enable.

```
SB_DFFESS_inst : SB_DFFESS
      port map (
      Q => Q,            -- Registered Output
      C => C,            -- Clock
      E => E,                  -- Clock Enable
      D => D,            -- Data
      S => S                   -- Synchronously Set
      );
```

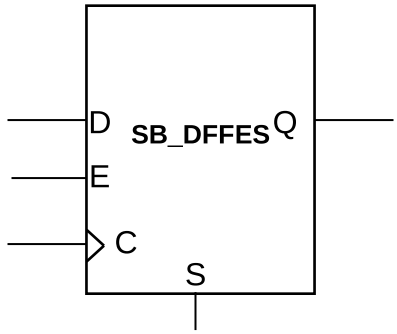
-- End of SB_DFFESS instantiation

SB_DFFES

D Flip-Flop with Clock Enable and Asynchronous Set

Data: D is loaded into the flip-flop when S is low and E is high during a rising clock edge transition.

Set: S input is active high, overrides all other inputs and asynchronously sets the Q output.



Inputs				Output
S	**E**	**D**	**CLK**	**Q**
1	X	X	X	1
0	0	X	X	Previous **Q**
0	1	0	↑	0
0	1	1	↑	1
Power on State	X	X	X	0

Key
↑ Rising Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage

This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values

The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'
Input S: Logic '0'
Input E: Logic '1'

Verilog Instantiation

// SB_DFFES - D Flip-Flop, Set is asynchronous on rising clock edge with Clock Enable.

```
SB_DFFES    SB_DFFES_inst (
      .Q(Q),            // Registered Output
      .C(C),            // Clock
      .E(E),            // Clock Enable
      .D(D),            // Data
      .S(S)             // Asynchronously Set
      );
```

// End of SB_DFFES instantiation

VHDL Instantiation

-- SB_DFFES - D Flip-Flop, Set is asynchronous on rising clock edge with Clock Enable.

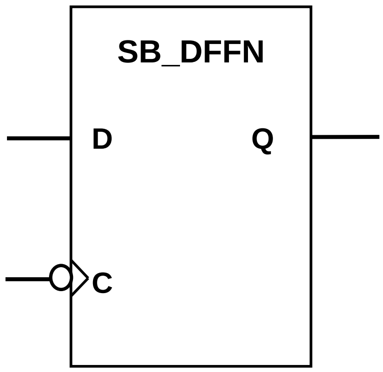
```
SB_DFFES_inst : SB_DFFES
      port map (
      Q => Q,           -- Registered Output
      C => C,           -- Clock
      E => E,                 -- Clock Enable
      D => D,           -- Data
      S => S                  -- Asynchronously Set
      );
```

-- End of SB_DFFES instantiation

SB_DFFN

D Flip-Flop – Negative Edge Clock

Data: D is loaded into the flip-flop during the falling clock edge transition.



Inputs			Output
	D	**C**	**Q**
	0	↘	0
	1	↘	1
Power on State	X	X	0

Key
↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage
This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values
The iCEcube2 software assigns the following signal values to unconnected input ports:

Input D: Logic '0'
Input C: Logic '0'

Verilog Instantiation

// SB_DFFN - D Flip-Flop – Negative Edge Clock.

```
SB_DFFN SB_DFFN_inst (
      .Q(Q),              // Registered Output
      .C(C),              // Clock
      .D(D),              // Data
      );
```

// End of SB_DFFN instantiation

VHDL Instantiation

-- SB_DFFN - D Flip-Flop – Negative Edge Clock.

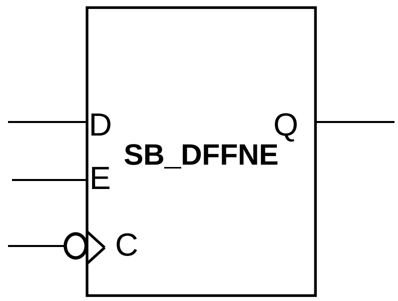
```
SB_DFFN_inst : SB_DFFN
      port map (
      Q => Q,            -- Registered Output
      C => C,            -- Clock
      D => D,            -- Data
      );
```

-- End of SB_DFFN instantiation

SB_DFFNE

D Flip-Flop – Negative Edge Clock and Clock Enable

Data: D is loaded into the flip-flop when E is high, during the falling clock edge transition.



Inputs			Output
E	**D**	**C**	**Q**
0	X	X	0
1	0	↘	0
1	1	↘	1
Power on State	X	X	0

Key
↘ Falling Edge
1 High logic level
0 Low logic level
X Don't care
? Unknown

HDL Usage
This register is inferred during synthesis and can also be explicitly instantiated.

Default Signal Values
The iCEcube2 software assigns the following signal values to unconnected input ports:
Input D: Logic '0'
Input C: Logic '0'
Input E: Logic '1'

Note that explicitly connecting a Logic '1' value to port E will result in a non-optimal implementation, since an extra LUT will be used to generate the Logic '1'. If the user's intention is to keep the FF always enabled, it is recommended that either port E be left unconnected, or the corresponding FF without a Clock Enable port be used.

Verilog Instantiation

// SB_DFFNE - D Flip-Flop – Negative Edge Clock and Clock Enable.

```
SB_DFFNE  SB_DFFNE_inst (
       .Q(Q),              // Registered Output
       .C(C),              // Clock
       .D(D),              // Data
       .E(E),              // Clock Enable
       );
```

// End of SB_DFFNE instantiation

VHDL Instantiation

-- SB_DFFNE - D Flip-Flop – Negative Edge Clock and Clock Enable.

```
SB_DFFNE_inst : SB_DFFNE
      port map (
      Q => Q,            -- Registered Output
      C => C,            -- Clock
      D => D,            -- Data
      E => E,                  -- Clock Enable
      );
```

-- End of SB_DFFNE instantiation